ROLL NO: _____

# MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

## END-SEM EXAMINATION — DECEMBER / JUNE 2024

FIRST YEAR (COMMON TO ALL BRANCHES) — SEMESTER 1

**Time Allowed: 3 Hours** **Maximum Marks: 70**

*NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.*
*2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].*
*3. Assume suitable missing data, if any, and state it clearly.*
*4. Use of scientific non-programmable calculator is permitted.*

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**--- UNIT I ---**

**Q.1** State and prove Rolle's Theorem. Expand e^x cos y using Maclaurin's Theorem. [14 Marks]

**[ OR ]**

**Q.1 (Alt)** State Leibniz's Rule for n-th derivative of product of two functions with proof. [14 Marks]

**--- UNIT II ---**

**Q.2** State Euler's Theorem for Homogeneous Functions. Find maximum and minimum of f(x,y)=x^3+y^3-3axy. [14 Marks]

**[ OR ]**

**Q.2 (Alt)** Evaluate Jacobian J=d(x,y,z)/d(r,theta,phi) for spherical polar coordinates. Lagrange multipliers. [14 Marks]

**--- UNIT III ---**

**Q.3** Evaluate double integral over region bounded by y=x^2 and y^2=x. Change order of integration. [14 Marks]

**[ OR ]**

**Q.3 (Alt)** Calculate volume of sphere using spherical polar triple integrals. [14 Marks]

**--- UNIT IV ---**

**Q.4** Find rank of matrix by reducing to Normal Echelon form. Solve system of linear equations. [14 Marks]

**[ OR ]**

**Q.4 (Alt)** State and prove Cayley-Hamilton Theorem. Diagonalize symmetric matrix using eigenvalues. [14 Marks]

**--- UNIT V ---**

**Q.5** Find Directional Derivative of scalar field. State and verify Green's Theorem in a plane. [14 Marks]

**[ OR ]**

**Q.5 (Alt)** State Stokes' Theorem and Gauss Divergence Theorem. Evaluate surface integrals over closed box. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****